

# Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference

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# Motivating problem

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# Effect of healthy food access on obesity rates

Poor healthy food access been associated with:

- high systolic blood pressure (Suarez et al., 2015)
- hypertension and cardiovascular disease (Kelli et al., 2019)
- obesity (Thomsen et al., 2016)

Challenges include:

- **Interference:** cross-border shopping
- **Heterogeneous effects:** magnitude and direction of effects can vary by community characteristics

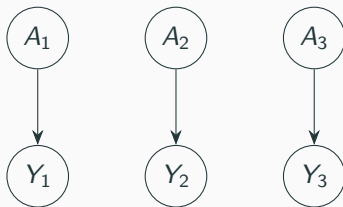
**Research question:** What is the effect of healthy food retailers on ZIP Code obesity rates and how does it vary for different ZIP Code characteristics?

# Causal inference with interference

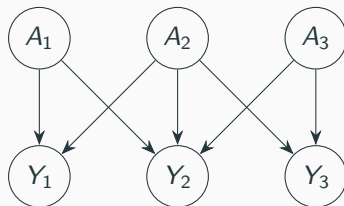
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# Interference

- **No interference assumption:** The treatment/exposure of one unit does not impact the outcome of another unit
- Often violated in studies with spatial treatments or exposures



**Figure 1:** No interference



**Figure 2:** Spatial network interference

# Notation and Potential outcomes

- $\mathbf{W}$  = adjacency matrix
  - $w_{ij}$  = strength of connection between units  $i$  and  $j$
  - if  $w_{ij} = 0$ , units  $i$  and  $j$  are unconnected
- $\mathcal{N}_i$  = set of units that are connected to unit  $i$ 
  - Call this set the neighbors of unit  $i$
- $(i, \mathcal{N}_i, \mathcal{N}_{-i})$  = partition of all units
- Potential outcome:  $Y(A_i, \tilde{A}_i = f(\mathbf{A}_{\mathcal{N}_i}))$ 
  - $(A_i, \tilde{A}_i)$  = joint treatment
  - $f(\mathbf{A}_{\mathcal{N}_i})$  is a summarizing function of the neighboring exposures/treatments

- **Direct effect (DE)**: effect of unit-level treatment ( $A$ ) on  $Y$  holding neighbor-level treatment fixed

$$DE(\tilde{A} = \tilde{a}) = E[Y(1, \tilde{a}) - Y(0, \tilde{a})]$$

- **Indirect (spillover) effect (IE)**: effect of neighbor-level treatment ( $\tilde{A}$ ) holding unit-level treatment fixed

$$IE(A = a) = E[Y(a, 1) - Y(a, 0)]$$

- For simplicity, we consider binary  $A$  and  $\tilde{A}$

# Assumptions

1. Consistency:  $Y_i = Y_i(\mathbf{A})$
2. **Neighborhood interference**: Given a function  $f$ , if  $f(\mathbf{A}_{\mathcal{N}_i}) = f(\mathbf{A}'_{\mathcal{N}_i})$ , then  $Y(A_i, \mathbf{A}_{\mathcal{N}_i}) = Y(A_i, \mathbf{A}'_{\mathcal{N}_i})$  (Forastiere et al., 2021; Hudgens and Halloran, 2008)
3. Unconfoundedness of the joint treatment:  
 $Y_i(a, \tilde{a}) \perp A_i, \tilde{A}_i | \mathbf{X} \quad \forall a, \tilde{a}, \quad \forall i$
4. Positivity:  $f(a, \tilde{a} | \mathbf{X}) > 0 \quad \forall \mathbf{x} \quad \text{s.t.} \quad f_{\mathbf{X}}(\mathbf{x}) > 0$

## Bayesian additive regression trees (BART)

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- Ensemble machine learning method that models the outcome as a sum of weak learners
- Bayesian framework: places priors on tree structure and parameters and provides uncertainty quantification via credible intervals
- Strong performance in causal inference problems (Hill, 2011; Dorie et al., 2022)

# Bayesian causal forests (BCF)

- Extension of BART designed for causal inference and heterogeneous treatment effects (Hahn et al., 2020)

$$Y_i = \mu(\mathbf{X}_i, \hat{\pi}(\mathbf{X}_i)) + \tau(\mathbf{X}_i)A_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- $\mu(\mathbf{X})$  = prognostic function
- $\tau(\mathbf{X})$  = moderating function
- Include estimate of the propensity score in the prognostic function

# Joint treatment Bayesian causal forests (jtBCF)

- Extension of Bayesian causal forests for joint treatments
- Separate moderating functions for both  $A$ ,  $\tilde{A}$ , and their interaction

$$Y_i = \mu(\mathbf{X}_i, \hat{\pi}_A, \hat{\pi}_{\tilde{A}}) + \tau_1(\mathbf{X}_i)A_i + \tau_2(\mathbf{X}_i)\tilde{A} + \tau_3(\mathbf{X}_i)A\tilde{A} + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

- $\mu(\mathbf{X})$ : prognostic effect
- $\tau_1(\mathbf{X})$ : direct effect (own treatment)
- $\tau_2(\mathbf{X})$ : indirect (spillover) effect
- $\tau_3(\mathbf{X})$ : interaction effect

- We use the `SoftBart` R package (Linero, 2022) to implement our model
  - `SoftBart` R package also includes functionality to develop custom BART models
- Soft BART replaces binary splits with smooth gating functions (e.g., logistic functions) so splits are probabilistic (Linero and Yang, 2018)

# Simulations

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# Simulation setup

- Use Texas ZIP Codes as set of units ( $n = 1927$ )
- Simulate three ZIP Code-level covariates and compute three neighborhood-level covariates, which are neighborhood means of the ZIP Code-level covariates
- Consider linear/nonlinear  $\mu(\mathbf{X})$
- Consider homogeneous/heterogeneous  $\tau_1(X) - \tau_3(X)$
- Treatment:

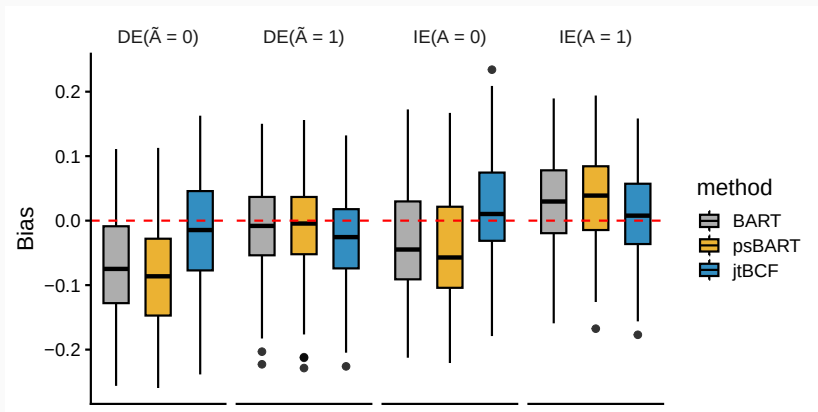
$$A \sim \text{Bernoulli}(0.8 \Phi(-1.25 + \mu(X)/\text{sd}(\mu(X)) + 0.5X_5) + u_i/10)$$

$$\tilde{A}_i = \mathbf{1}(\bar{A}_{\mathcal{N}_i} > 0.5)$$

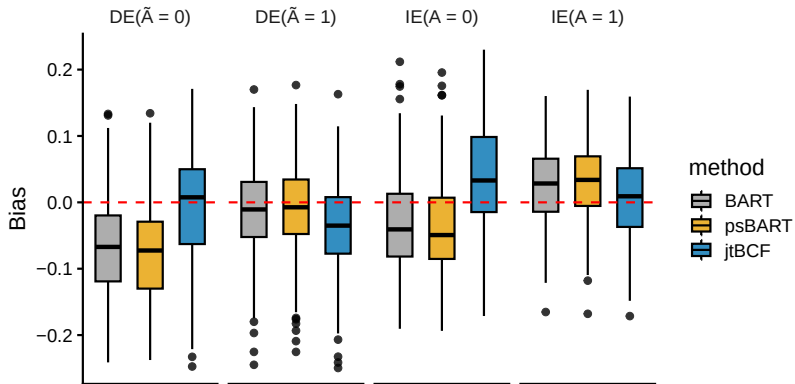
- Outcome:

$$y \sim N(\mu(X) + \tau_1(X)A + \tau_2(X)\tilde{A} + \tau_3(X)A\tilde{A}, 1)$$

## Simulation results (Nonlinear/Homogeneous case)



## Simulation results (Nonlinear/Heterogeneous case)



## Coverage - Direct and Indirect effects

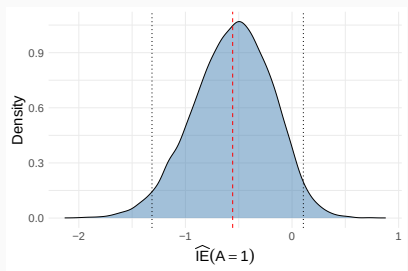
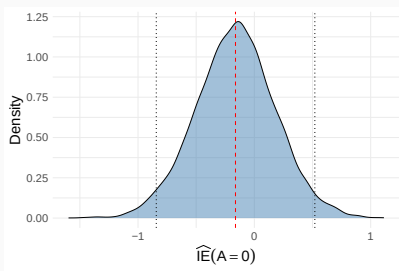
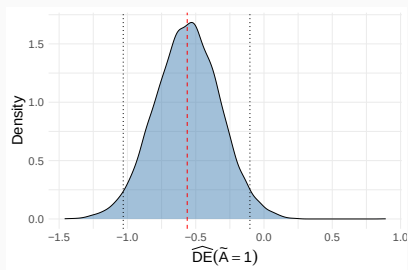
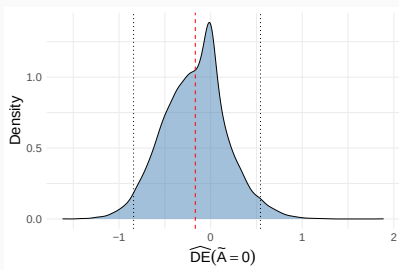
| Setting       | Estimand            | Soft BART | Soft psBART | Soft jtBCF |
|---------------|---------------------|-----------|-------------|------------|
| <b>Homo</b>   |                     |           |             |            |
|               | $DE(\tilde{A} = 0)$ | 0.80      | 0.78        | 0.94       |
|               | $DE(\tilde{A} = 1)$ | 0.94      | 0.94        | 0.93       |
|               | $IE(A = 0)$         | 0.90      | 0.86        | 0.93       |
|               | $IE(A = 1)$         | 0.91      | 0.90        | 0.96       |
| <b>Hetero</b> |                     |           |             |            |
|               | $DE(\tilde{A} = 0)$ | 0.82      | 0.82        | 0.96       |
|               | $DE(\tilde{A} = 1)$ | 0.92      | 0.93        | 0.93       |
|               | $IE(A = 0)$         | 0.93      | 0.93        | 0.92       |
|               | $IE(A = 1)$         | 0.95      | 0.95        | 0.97       |

## **Case study: Effect of healthy food access on obesity rates in Texas**

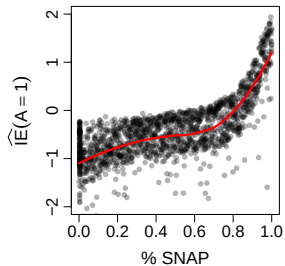
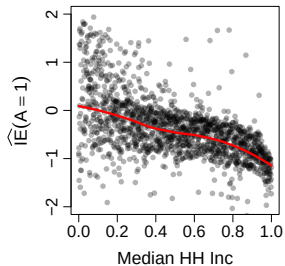
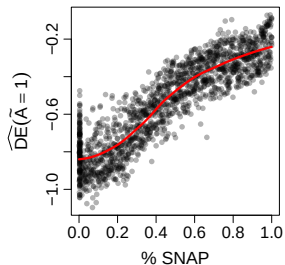
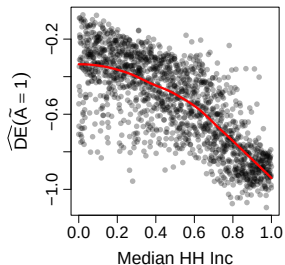
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- **Unit of analysis:** Texas ZIP Codes in 2022
- **Outcome:** Obesity prevalence ( $y \in (0, 100)$ )
  - Model-based estimate adjusted for age, sex, race, and education
- **Exposures:**
  - $A$ : Indicator for  $\geq 1$  healthy food retailer in the ZIP Code.
  - $\tilde{A}$ : Indicator for  $\geq 1$  healthy food retailer in neighboring ZIP Codes.
- **Covariates:** Median household income, % SNAP, unemployment, % commuting by public transit, NDVI, % married, % veterans, and % with disability
- **Neighborhood:** Informed by driving times between ZIP Code centroids

# Posterior distributions of direct/indirect effects



# Results: Heterogeneous effects



# Discussion

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# Conclusions

- jtBCF model offers strong coverage and bias reduction in estimating direct and indirect effects compared to BART and psBART
- Generalizable to any setting with joint treatments
- Limitations:
  - Model has increased variability due to noise coming from three separate moderating functions
  - Motivating example has some weaknesses:
    - Data is cross-sectional
    - Obesity rate at ZIP Code level is model-based estimate (takes into account race, sex, age, and education)
    - Food retailer list is not exhaustive; does not include farmer's markets or food banks

# Thank you!

## Coauthors

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- Dr. Shariq Mohammed (BU Biostats)
- Dr. Sara Lodi (BU Biostats)

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## Any questions?



**QR Code for Slides**

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